AGILE LAB FAT

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EMPLOYEE PAYROLL SYSTEM

1. PAYROLL LOGIC :

Objective:

Calculate **employee net salary** based on:

- Basic Salary
- Allowances (HRA, DA, Bonus)
- Deductions

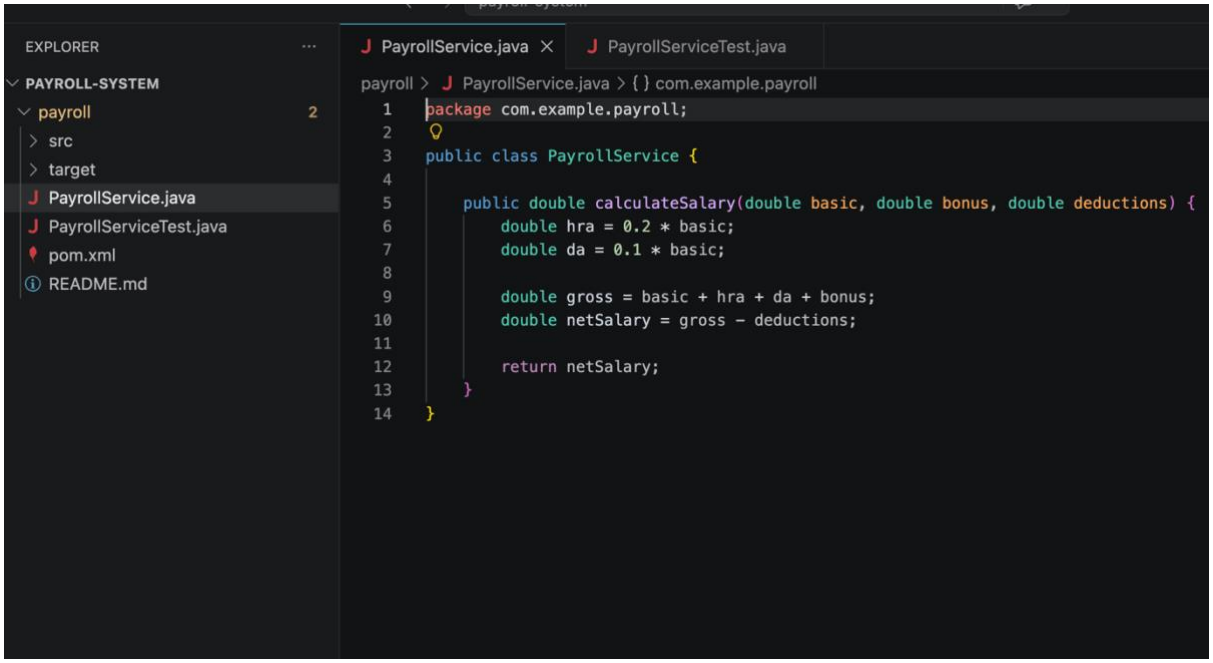
Formula used :

Gross Salary = Basic + HRA + DA + Bonus

Net Salary = Gross Salary – Deductions

2.Junit test : (Java program Development)

Initially we have to get code for normal **payrollservie.java**



Then Junit test for **PayrollServiceTest.java**

EXPLORER

PAYROLL-SYSTEM

- payroll 2
  - src
  - target
  - PayrollService.java
  - PayrollServiceTest.java
  - pom.xml
  - README.md

PayrollService.java | PayrollServiceTest.java

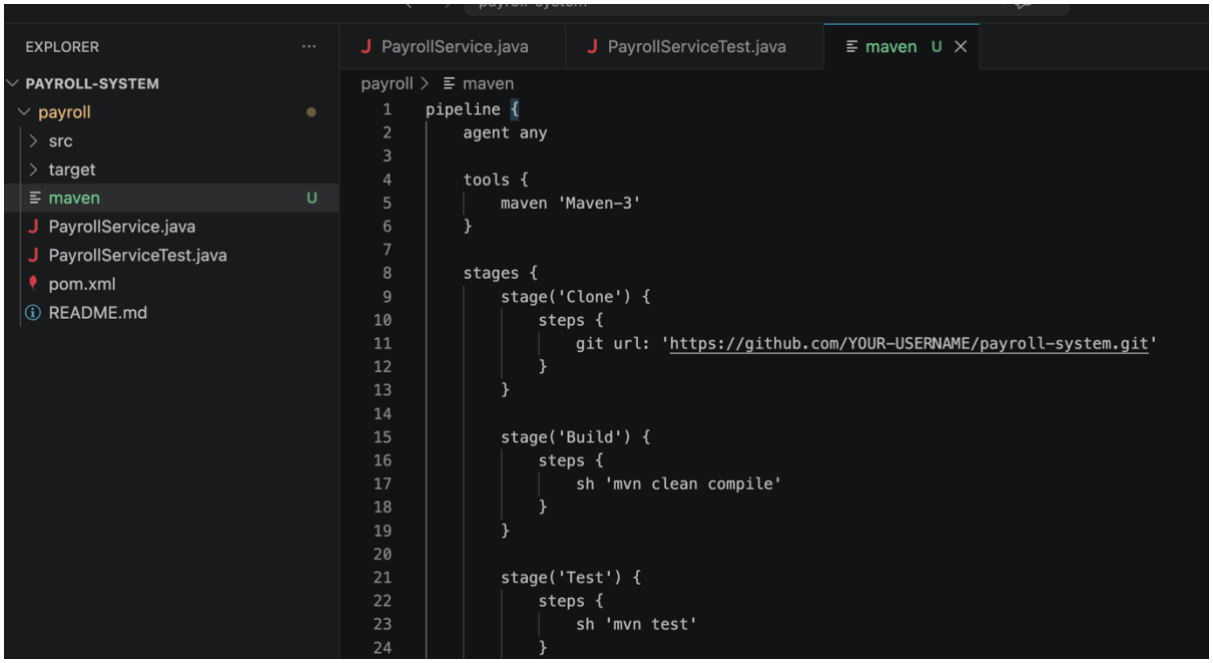
payroll > PayrollServiceTest.java

```java
package com.example.payroll;

import org.junit.Test;
import static org.junit.Assert.*;

public class PayrollServiceTest {

    PayrollService service = new PayrollService();

    @Test
    public void testSalaryCalculation() {
        double salary = service.calculateSalary(10000, 2000, 1000);
        assertEquals(13000.0, salary, 0.1);
    }

    @Test
    public void testNoBonus() {
        double salary = service.calculateSalary(10000, 0, 1000);
        assertEquals(11000.0, salary, 0.1);
    }
}
```

JUNIT TESTING :

```
Last login: Wed Apr 15 14:02:55 on ttys000
(base) rishi_2006@RISHIs-MacBook-Air ~ % cd Lab Fat
cd: string not in pwd: Lab
(base) rishi_2006@RISHIs-MacBook-Air ~ % cd Labfat
cd: no such file or directory: Labfat
(base) rishi_2006@RISHIs-MacBook-Air ~ % cd desktop
(base) rishi_2006@RISHIs-MacBook-Air desktop % cd Labfat
(base) rishi_2006@RISHIs-MacBook-Air Labfat % mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.wallet:wallet-app >-----------------------
[INFO] Building wallet-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ wallet-app ---
[INFO] Deleting /Users/rishi_2006/Desktop/Labfat/target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ wallet-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/rishi_2006/Desktop/Labfat/src/main/resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ wallet-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ wallet-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/rishi_2006/Desktop/Labfat/src/test/resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ wallet-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[INFO]
[INFO] --- surefire:3.0.0-M9:test (default-test) @ wallet-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.wallet.WalletTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.017 s - in com.wallet.WalletTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.008 s
[INFO] Finished at: 2026-04-15T18:10:04+05:30
[INFO] ------------------------------------------------------------------------
(base) rishi_2006@RISHIs-MacBook-Air Labfat %
```

3.Implement CI/CD pipleline :



Created :

Github setup

Configure pipeline

Build maven

Done with code of pipeline and build successfully

localhost

Personal Access Tokens (Classic) | payroll-project Config - Jenkins

Jenkins / payroll-project / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

try sample Pipeline...

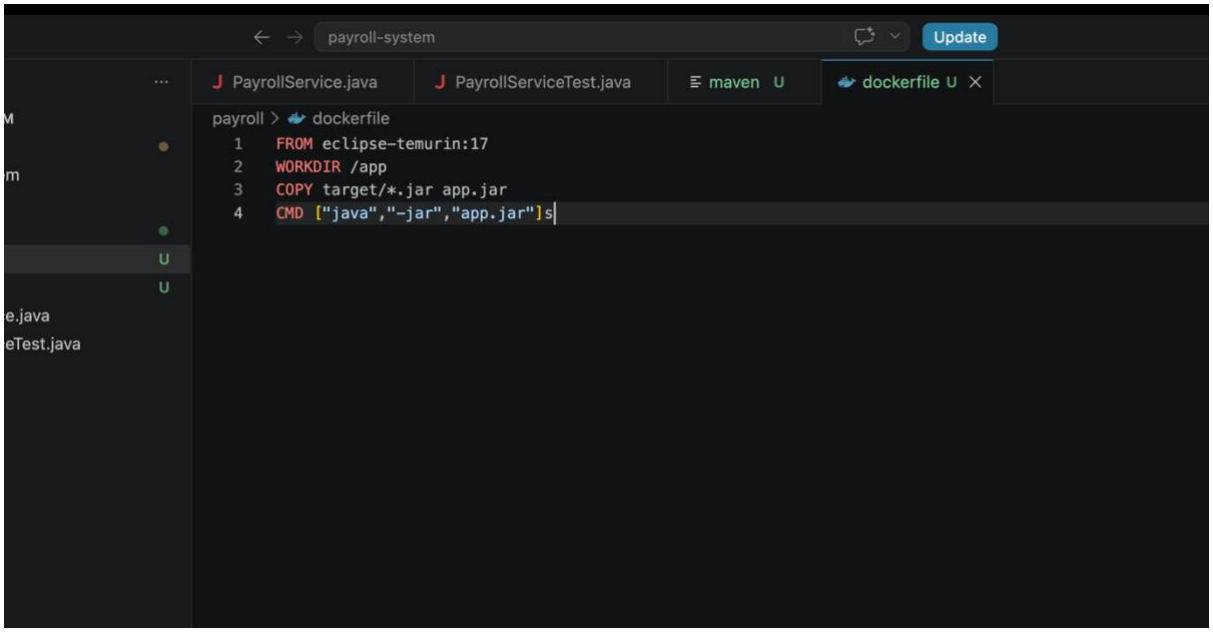
```
pipeline {
    agent any

    tools {
        maven 'Maven-3'
    }

    stages {
        stage('Clone') {
            steps {
                git url: 'https://github.com/rangeshrmr2023-oss/payrollproject.git'
            }
        }

        stage('Build') {
```

Use Groovy Sandbox ?

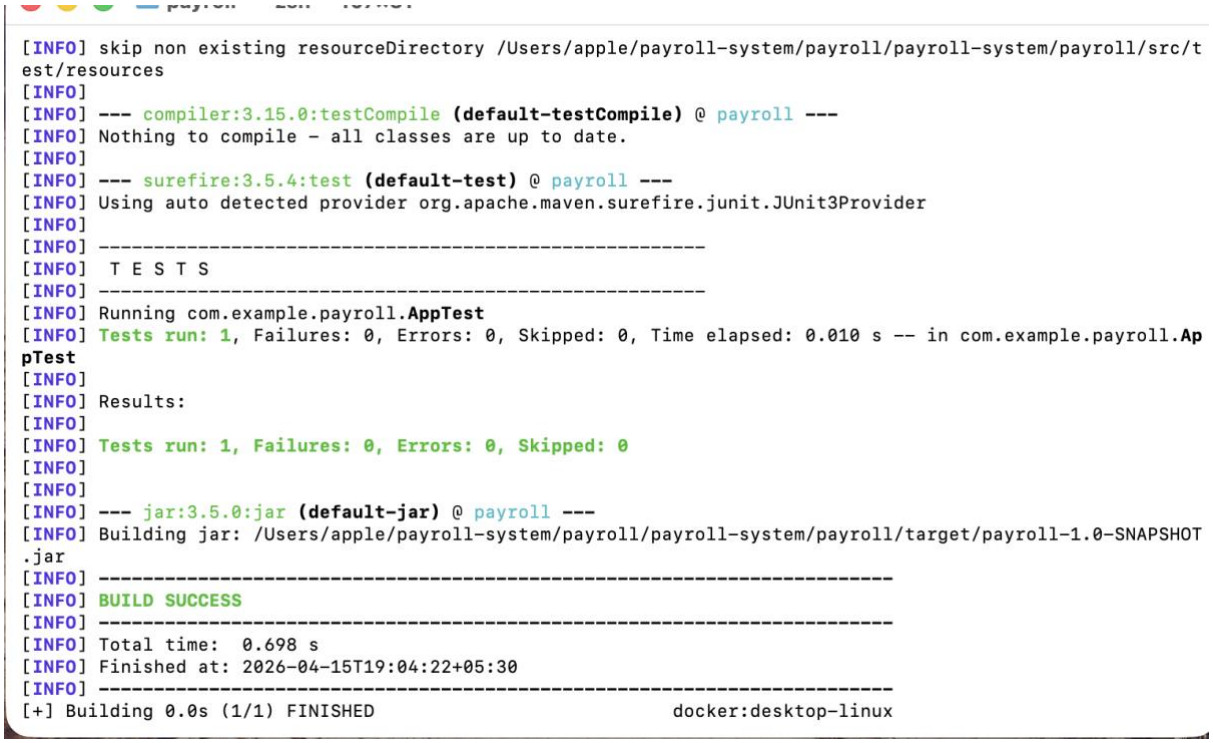
Save | Apply

deploy service on kuberntes and docker:

Docker setup :



Docker running successfully :



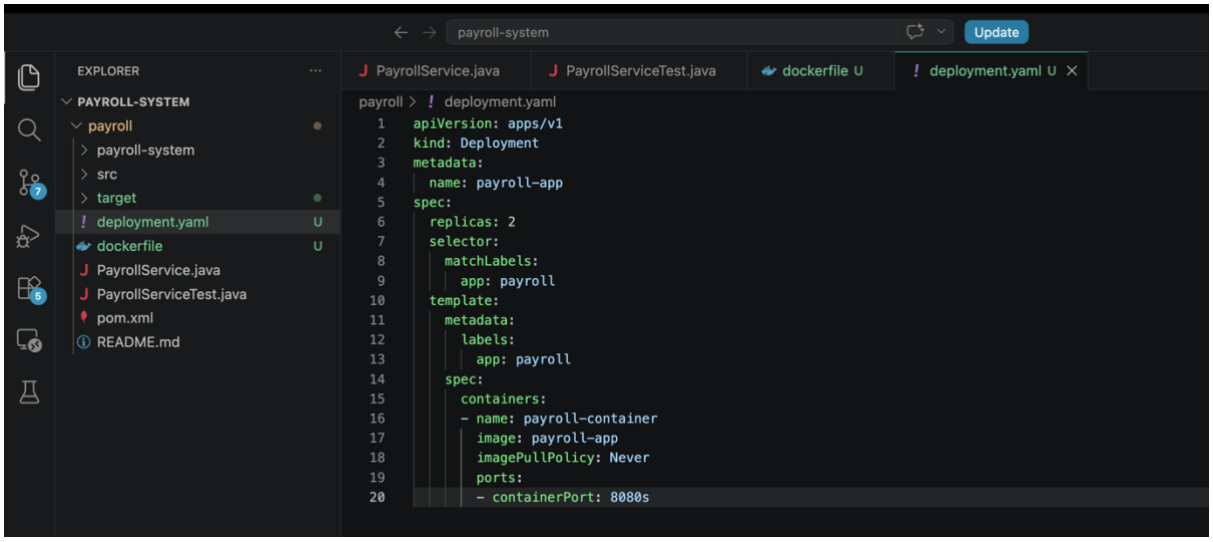
```
[INFO] skip non existing resourceDirectory /Users/apple/payroll-system/payroll/payroll-system/payroll/src/t
est/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ payroll ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payroll ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.payroll.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.010 s -- in com.example.payroll.Ap
pTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ payroll ---
[INFO] Building jar: /Users/apple/payroll-system/payroll/payroll-system/payroll/target/payroll-1.0-SNAPSHOT
.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.698 s
[INFO] Finished at: 2026-04-15T19:04:22+05:30
[INFO] ------------------------------------------------------------------------
[+] Building 0.0s (1/1) FINISHED                          docker:desktop-linux
```

Docker pushed :

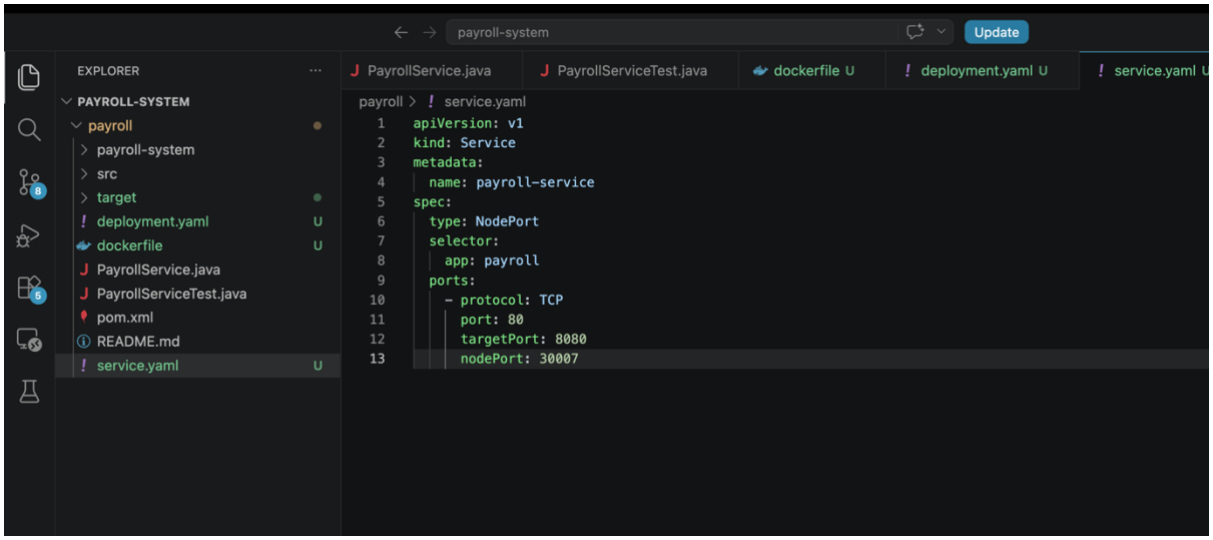
```
05de165e5c77: Pushed
dd1ff60268d3: Pushed
ca09713b1606: Pushed
95691bedf2b2: Pushed
46445512c2e6: Pushed
4e87bccff4c7: Pushed
c6d202f2689b: Pushed
a1f6fc2d8b91: Pushed
latest: digest: sha256:28a6931c6b5e4dfc1f19ad9cc4c976f534e2e361478d3d56ef1dfae9278f8eb9 size: 856
```

Creating Kubernetes Deployment and service Yaml files:

Deployment.yaml



Service.yaml




```
05de165e5c77: Pushed
dd1ff60268d3: Pushed
ca09713b1606: Pushed
95691bedf2b2: Pushed
46445512c2e6: Pushed
4e87bccff4c7: Pushed
c6d202f2689b: Pushed
a1f6fc2d8b91: Pushed
latest: digest: sha256:28a6931c6b5e4dfc1f19ad9cc4c976f534e2e361478d3d56ef1dfae9278f8eb9 size: 856
(base) rishi_2006@RISHIs-MacBook-Air Labfat % kubectl apply -f k8s-manifest.yaml
```